

The background is a light blue gradient with various digital and web-related icons and text elements. At the top, there are labels for 'HTML', 'CSS', and 'JS' in different colors. A green starburst icon is on the left. Below it, a laptop screen displays 'WEB DESIGN' and some code. To the right of the laptop, there's a small line graph. At the bottom, there's a waveform icon. The overall theme is web development and programming.

Programación del Lado del Servidor

Unidad IV

Programación del Lado del Servidor

4.1 Introducción al Lenguaje

4.2 Estructuras de Control

4.3 Tratamiento de Formularios

4.4 Manejo de Objetos del Servidor

4.5 Creación de Clases

4.6 Acceso a Datos

Practica Unidad 4: Programación del Lado del Servidor

- Introducción:

- Servidores Web
- HTML
- CSS
- Validador Oficial
 - HTML: <https://validator.w3.org/>
 - CSS: <https://jigsaw.w3.org/css-validator/>
- GitHub
- JavaScript
- Librerías
- Frameworks
- Programación del lado del Servidor
- Base de Datos



Practica Unidad 4: Programación del Lado del Servidor

- Objetivo:
 - Conocer y aplicar un lenguaje de programación del lado del servidor.
 - Construir aplicaciones Web dinámicas usando conectividad a orígenes de datos (BD)



Practica Unidad 4: Programación del Lado del Servidor

- Material y Equipo:
 - Laptop o computadora de escritorio
 - Servidores Web
 - Manejador de Base de Datos
 - Validadores oficiales
 - GitHub
 - JavaScript
 - Librerías
 - Frameworks
 - Programación del lado del Servidor:
 - JSP
 - Django
 - Node.js
 - Php



Practica Unidad 4: Programación del Lado del Servidor

- Descripción de la Problemática:
 - Basado en la problemática anterior, la empresa DevWeb necesita almacenar y gestionar todos los datos de la pagina Web en una BD así como implementar algunas medidas de seguridad.



Programación Web -- ISC Rubén Lara

Practica Unidad 4: Programación del Lado del Servidor

- Paso 1:
 - Crear una clase o plantilla para la conexión a la BD
 - Mostrar todos los productos y servicios almacenados en la BD, manejar eventos y diseño
- Paso 2:
 - Crear un modulo para registrar, modificar y eliminar usuarios de la Pagina Web
 - Usar permisos, roles y tipos de usuario correctos en la BD
 - Cifrar contraseñas de los usuarios en la BD
- Paso 3:
 - De la practica anterior en la plantilla de formularios, almacenar y actualizar la informacion en la BD
 - Usar AJAX para minimo un elemento
 - Nombre de Usuario
 - Nombre / Id de Producto o Servicio
 - Conservar Sesiones del Navegador
- Paso 4:
 - Seguir validando el codigo fuente HTML de todas las paginas renderizadas.
 - Seguir validando los CSSs.
 - El Servidor Web debe tener conexion cifrada (HTTPS)
 - Commits en GitHub

Practica Unidad 4: Programación del Lado del Servidor

- Paso Opcional (Hotel):
 - Crear modulo que permita recibir pagos de reservaciones.

API	Dificultad de Integración	Métodos Principales	Sandbox (Pruebas)	Comentarios
Mercado Pago (Checkout Pro)	★ (Muy Baja)	Tarjetas, Oxxo, SPEI, Saldo	✓ Sí (Excelente)	Para Principiantes, integración directa con Oxxo, Modulo de pruebas ideal
PayPal (Smart Buttons)	★★ (Baja)	Tarjetas, Saldo PayPal	✓ Sí	Dificultad baja, uso directo con JavaScript
Clip (Checkout / Links)	★ (Muy Baja)	Tarjetas, Puntos	✓ Sí	Mas rápido de implementar
Conekta	★★★ (Media)	Oxxo, Tarjetas, SPEI	✓ Sí	Directo para Backend, convenio con Oxxo
Stripe (Checkout)	★★★★ (Media/Alta)	Tarjetas, OXXO	✓ Sí	Dificultad alta, uso de Webhooks

Practica Unidad 4: Programación del Lado del Servidor

- Entregables:
 - PDF con la siguiente información:
 - Diseño de red con direcciones IP, servicios y datos de acceso
 - Incluir en orden los 4 pasos de la practica
 - Breve descripción del procedimiento
 - Comandos y/o configuración realizada
 - Captura de pantalla completa
 - Conclusiones
 - Código Fuente
 - Revisión
 - Opción 1: en el salón/laboratorio
 - Opción 2: realizar un video de 10 minutos narrando los 4 pasos de la practica.

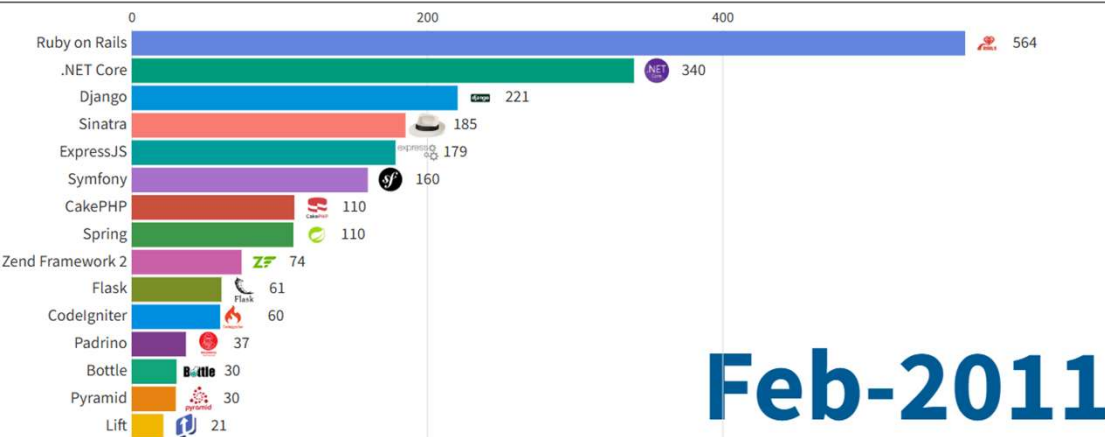
Practica Unidad 4: Programación del Lado del Servidor

- Evaluación:
 - En parejas
 - Derecho a Evaluación solo para esta Unidad 4:
 - Opción 1:
 - Tutorial Lenguaje del lado del Servidor: todas las lecciones, todos los ejercicios, examen
 - Opción 2:
 - Entregar todas las actividades de la Unidad
 - Fecha de entrega:
 - 4 al 6 de mayo: 100
 - 7 y 8 de mayo: 80
 - Por cada respuesta o elemento faltante se restaran 10 puntos
 - Puntos extra por usar elementos diferentes a los ejemplos de clase
 - Todos los archivos (PDF, video) deben tener el nombre con la siguiente estructura:
 - *AP1#CONTROL_AP2#CONTROL_U4_PW_G.EXTENSION*

4.1 Introducción al Lenguaje

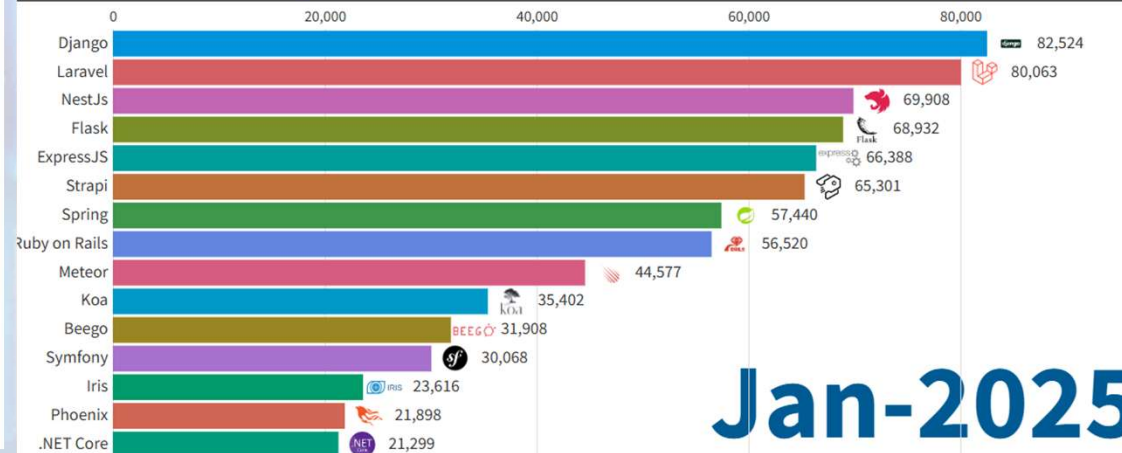
- Cual lenguaje elegir?
- <https://statisticsanddata.org/data/most-popular-backend-frameworks-2012-2025/>

Most Popular Backend Frameworks



Feb-2011

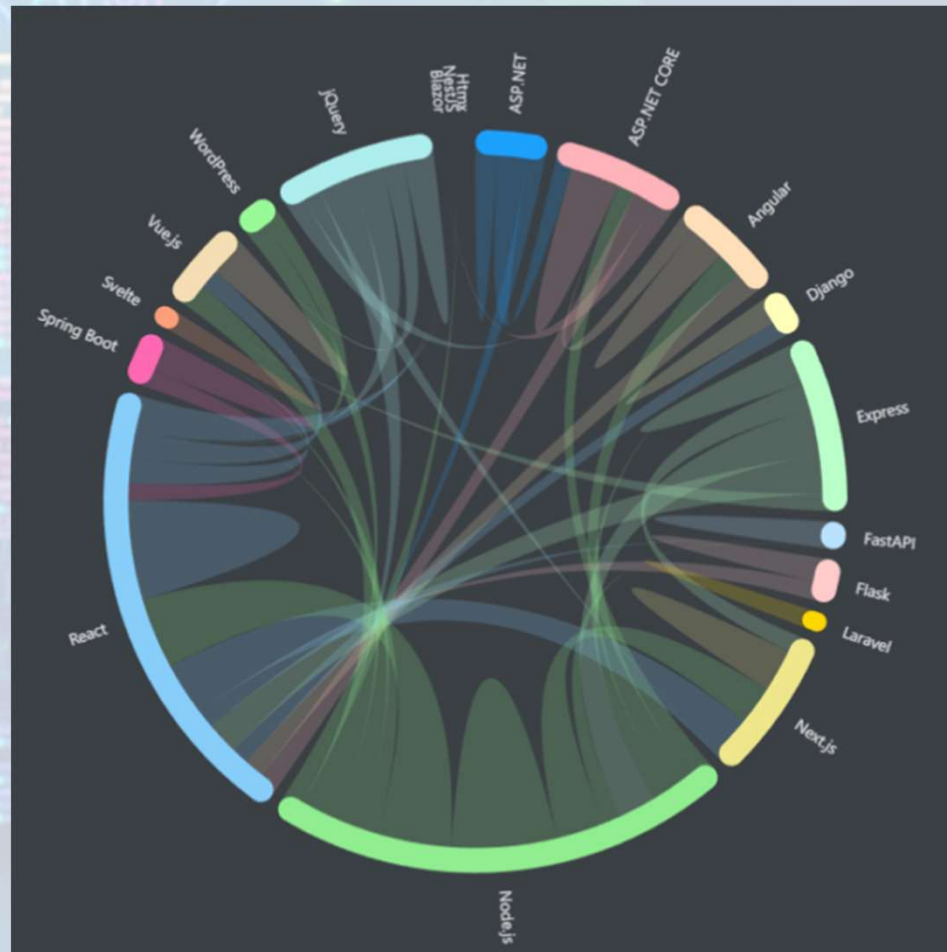
Most Popular Backend Frameworks



Jan-2025

4.1 Introducción al Lenguaje

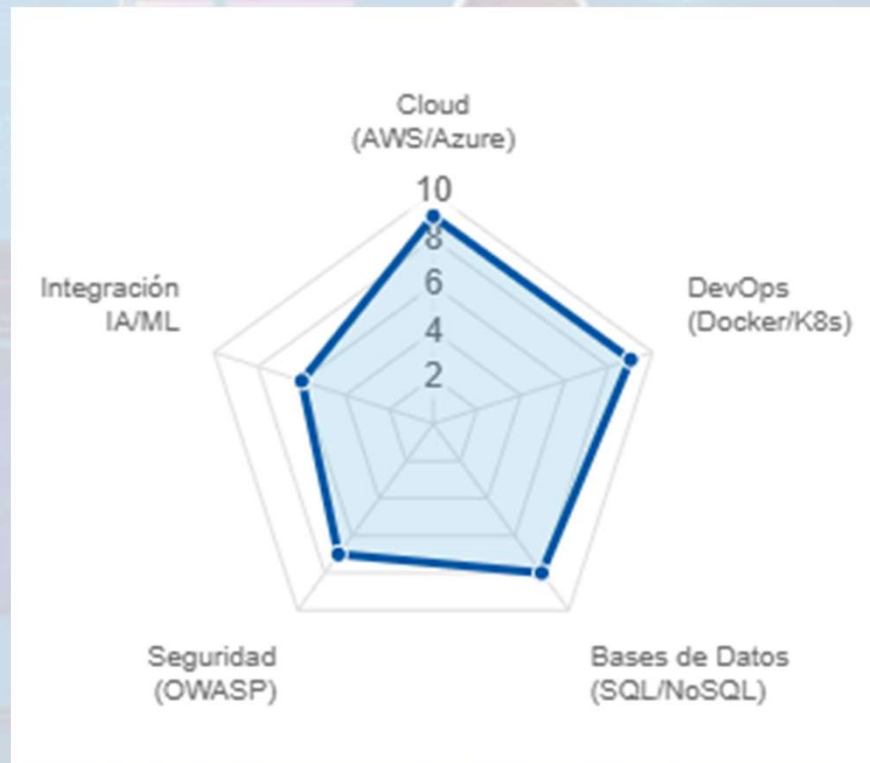
- Cambios de Tecnología:



Programación Web -- ISC Rubén Lara

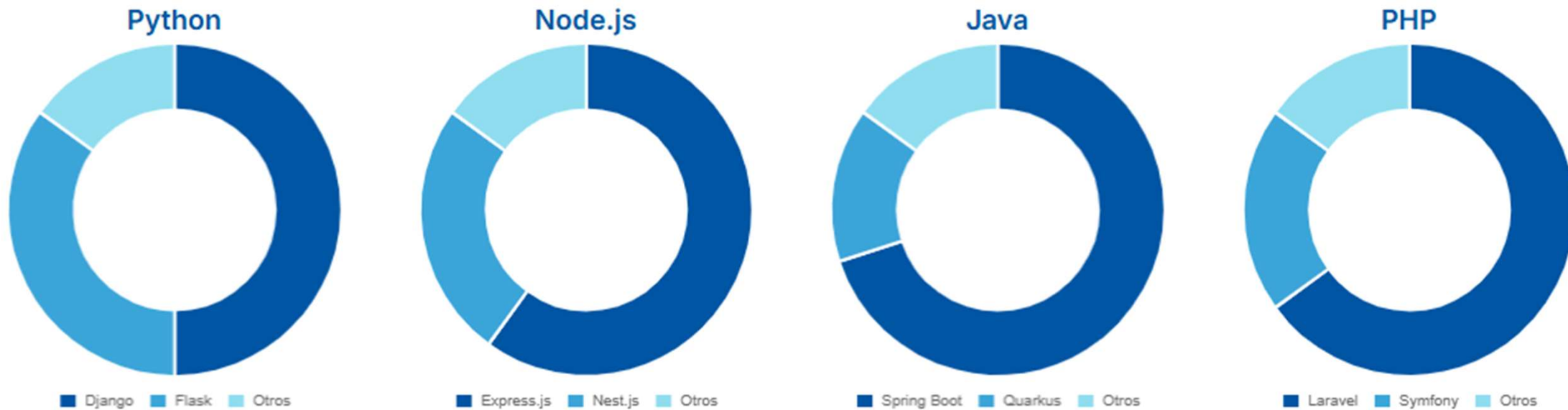
4.1 Introducción al Lenguaje

- Habilidades Clave:



4.1 Introducción al Lenguaje

- Frameworks:



4.1 Introducción al Lenguaje

- Tecnologías Emergentes:

- Go (Golang)

- Futuro Cloud Native
 - Microservicios
 - Alta concurrencia

- Rust

- Seguridad
 - Reemplazo de C/C++
 - Blockchain
 - Rendimiento critico



4.1 Introducción al Lenguaje

- Django
 - Framework de Python para creación de sitios Web
 - Reutilización de componentes DRY (Don't Repeat Yourself)
 - Plantillas de:
 - Inicio de sesión
 - Conexión a BD
 - Operaciones CRUD (Create Read Update and Delete)

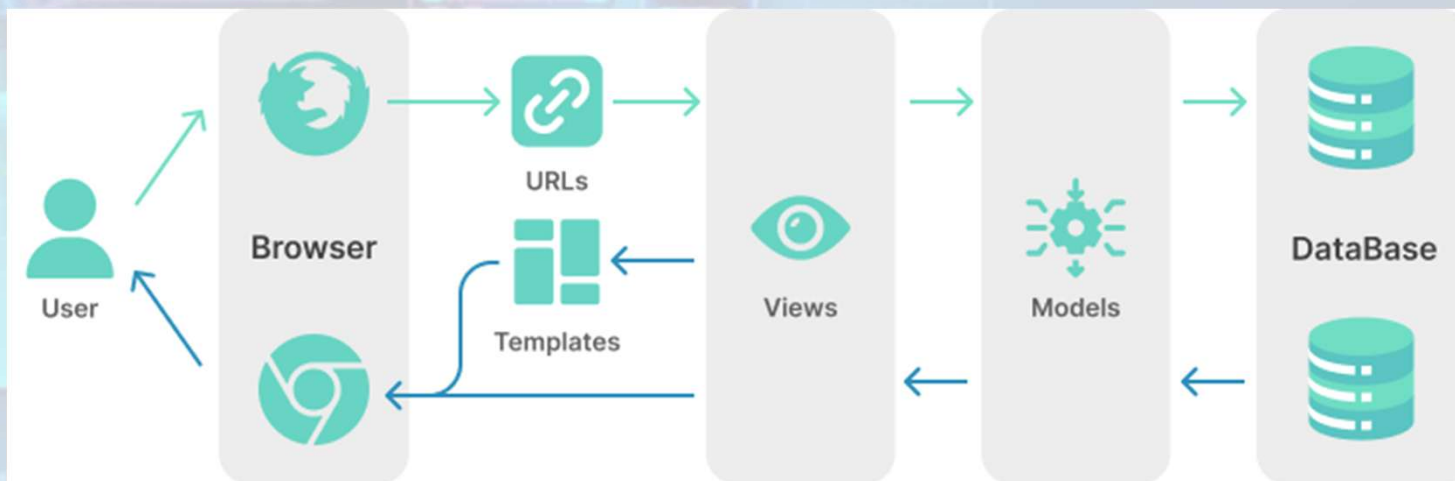


Programación Web -- ISC Rubén Lara



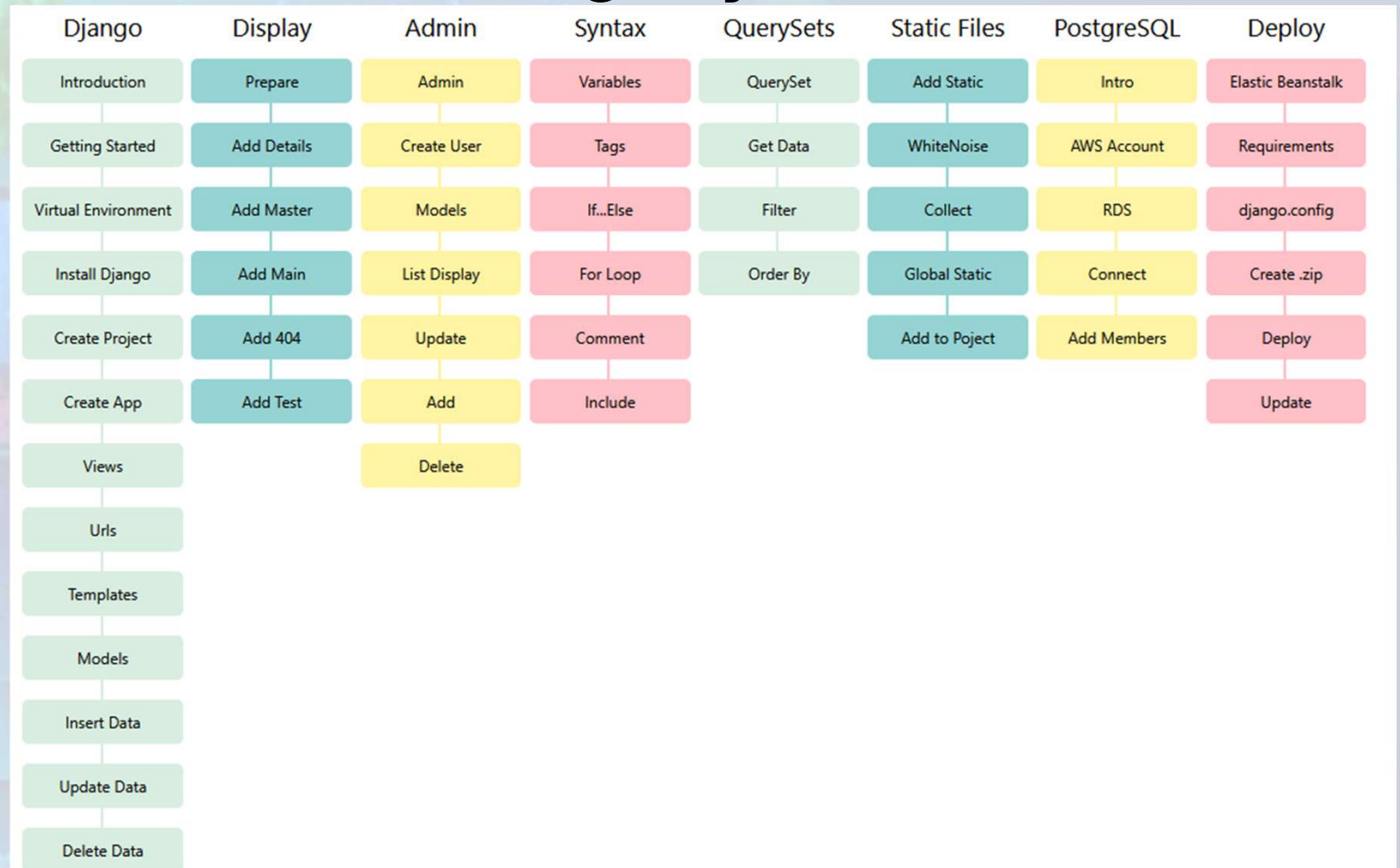
4.1 Introducción al Lenguaje

- Django
 - Utiliza el patrón de diseño MVT (Model View Template)
 - Modelo: Datos (Origen de Datos) *models.py*
 - Vista: Solicitudes y Contenido *views.py*
 - Plantilla: HTML con el diseño de la pagina Web, lógica y despliegue de datos *templates*
 - URLs: *urls.py*



4.1 Introducción al Lenguaje

- Django
- Proyecto:



4.1 Introducción al Lenguaje

- Django
 - Instalación
 - Tener Python instalado
 - Crear un Entorno Virtual para cada proyecto de Django

```
C:\Users\ruben>w:

W:\>cd "TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS"

W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS>python --version
Python 3.13.1

W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS>pip --version
pip 25.1.1 from V:\PY\Lib\site-packages\pip (python 3.13)

W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS>python -m venv pw2026-1-A

W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS>pw2026-1-A\Scripts\activate.bat

(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS>
```

```
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS>python -m pip install Django
Collecting Django
  WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=None)) a
tTimeoutError(<pip._vendor.urllib3.connection.HTTPSConnection object at 0x000001CED3CE02D0>
ed.org timed out. (connect timeout=15)'): /packages/e9/47/3d61d611609764aa71a37f7037b870e7
go-6.0.4-py3-none-any.whl.metadata
  WARNING: Retrying (Retry(total=3, connect=None, read=None, redirect=None, status=None)) a
tTimeoutError(<pip._vendor.urllib3.connection.HTTPSConnection object at 0x000001CED3CE0410>
ed.org timed out. (connect timeout=15)'): /packages/e9/47/3d61d611609764aa71a37f7037b870e7
go-6.0.4-py3-none-any.whl.metadata
  Downloading django-6.0.4-py3-none-any.whl.metadata (3.9 kB)
Collecting asgiref<=3.9.1 (from Django)
  Downloading asgiref-3.11.1-py3-none-any.whl.metadata (9.3 kB)
Collecting sqlparse<=0.5.0 (from Django)
  Downloading sqlparse-0.5.5-py3-none-any.whl.metadata (4.7 kB)
Collecting tzdata (from Django)
  Downloading tzdata-2026.1-py2.py3-none-any.whl.metadata (1.4 kB)
  Downloading django-6.0.4-py3-none-any.whl (8.4 MB)
8.4/8.4 MB 10.7 MB/s eta 0:00:00
  Downloading asgiref-3.11.1-py3-none-any.whl (24 kB)
  Downloading sqlparse-0.5.5-py3-none-any.whl (46 kB)
  Downloading tzdata-2026.1-py2.py3-none-any.whl (348 kB)
Installing collected packages: tzdata, sqlparse, asgiref, Django
Successfully installed Django-6.0.4 asgiref-3.11.1 sqlparse-0.5.5 tzdata-2026.1

[notice] A new release of pip is available: 24.3.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS>django-admin --version
6.0.4
```

```
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS>
```

```
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS>dir pw2026-1-A
El volumen de la unidad W es DAT2
El número de serie del volumen es: 26B2-FE71

Directorio de W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A

15/04/2026 05:40 p. m. <DIR>      .
15/04/2026 05:40 p. m. <DIR>      ..
15/04/2026 05:40 p. m.           71 .gitignore
15/04/2026 05:40 p. m.           Include
15/04/2026 05:40 p. m. <DIR>      Lib
15/04/2026 05:40 p. m.           194 pyvenv.cfg
15/04/2026 05:43 p. m. <DIR>      Scripts
                                2 archivos      265 bytes
                                5 dirs  33,901,789,184 bytes libres
```

4.1 Introducción al Lenguaje

- Django
- Proyectos

```
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A>django-admin startproject pw
```

```
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A>dir pw
El volumen de la unidad W es DAT2
El número de serie del volumen es: 26B2-FE71
```

```
Directorio de W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A\pw
```

```
15/04/2026 05:49 p. m. <DIR> .
15/04/2026 05:49 p. m. <DIR> ..
15/04/2026 05:49 p. m.      680 manage.py
15/04/2026 05:49 p. m. <DIR> pw
                1 archivos      680 bytes
                3 dirs  33,895,395,328 bytes libres
```

```
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A>dir pw\pw
El volumen de la unidad W es DAT2
El número de serie del volumen es: 26B2-FE71
```

```
Directorio de W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A\pw\pw
```

```
15/04/2026 05:49 p. m. <DIR> .
15/04/2026 05:49 p. m. <DIR> ..
15/04/2026 05:49 p. m.      397 asgi.py
15/04/2026 05:49 p. m.    3,105 settings.py
15/04/2026 05:49 p. m.      780 urls.py
15/04/2026 05:49 p. m.      397 wsgi.py
15/04/2026 05:49 p. m.         0 __init__.py
                5 archivos     4,679 bytes
                2 dirs  33,895,395,328 bytes libres
```

```
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A>python pw\manage.py runserver
Watching for file changes with StatReloader
Performing system checks...
```

```
System check identified no issues (0 silenced).
```

```
You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, aut
h, contenttypes, sessions.
```

```
Run 'python manage.py migrate' to apply them.
```

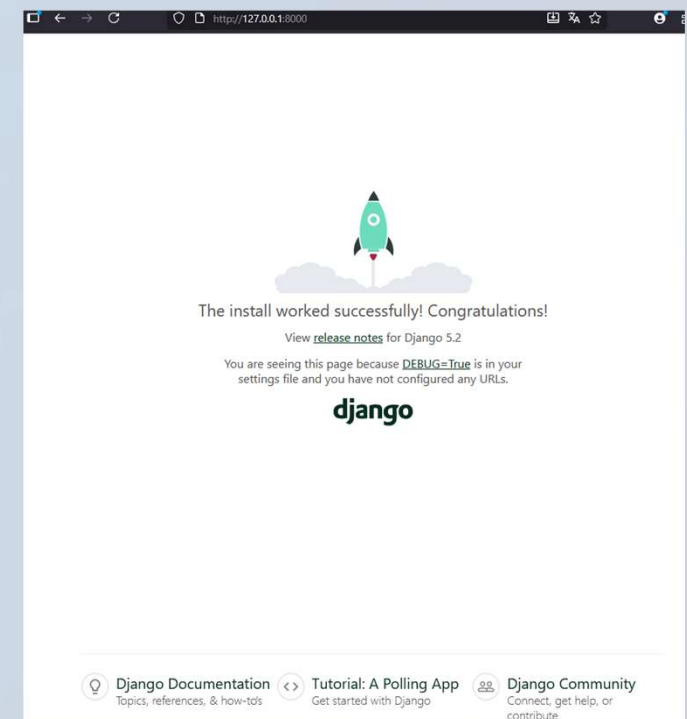
```
April 15, 2026 - 17:52:01
```

```
Django version 6.0.4, using settings 'pw.settings'
```

```
Starting development server at http://127.0.0.1:8000/
```

```
Quit the server with CTRL-BREAK.
```

```
WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/6.0/howto/deployment/
```



4.1 Introducción al Lenguaje

- Django
 - Apps
 - Uso específico dentro del proyecto
 - Inicio
 - Formularios

```
W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A>Scripts\activate.bat
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A>cd pw
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A\pw>python manage.py startapp inicio
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A\pw>dir inicio
El volumen de la unidad W es DAT2
El número de serie del volumen es: 26B2-FE71

Directorio de W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A\pw\inicio

15/04/2026  05:59 p. m.      <DIR>          .
15/04/2026  05:59 p. m.      <DIR>          ..
15/04/2026  05:59 p. m.                66 admin.py
15/04/2026  05:59 p. m.                92 apps.py
15/04/2026  05:59 p. m.      <DIR>          migrations
15/04/2026  05:59 p. m.                60 models.py
15/04/2026  05:59 p. m.                63 tests.py
15/04/2026  05:59 p. m.                66 views.py
15/04/2026  05:59 p. m.                 0 __init__.py
               6 archivos                347 bytes
               3 dirs 33,895,378,944 bytes libres
```

4.1 Introducción al Lenguaje

- Django
 - Visual Studio Code
 - Extensiones
 - Python – Microsoft
 - Django – Baptiste Darthenay
 - Django Templates – Bibhasdn



Python

Microsoft microsof.com | 213,473,837 | ★★★★★ (628)

Python language support with extension access points for IntelliSense (Pylance), Debugging (Python Debugger), linting, formatting, refactoring, unit tests, and more.

[Switch to Pre-Release Version](#) [Disable](#) [Uninstall](#) [Auto Update](#)

[DETAILS](#) [FEATURES](#) [CHANGELOG](#) [EXTENSION PACK](#)

Python extension for Visual Studio Code

A [Visual Studio Code extension](#) with rich support for the [Python language](#) (for all [actively supported Python versions](#)), providing access points for extensions to seamlessly integrate and offer support for IntelliSense (Pylance), debugging (Python Debugger), formatting, linting, code navigation, refactoring, variable explorer, test explorer, environment management (**NEW** Python Environments Extension).

Support for vscod.dev

The Python extension does offer [some support](#) when running on vscod.dev (which includes github.dev). This includes partial IntelliSense for open files in the editor.

Installation

Identifier	ms-python.python
Version	2026.4.0
Last Updated	2 minutes ago
Size	30.42MB
Cache	1.09KB

Marketplace

Published	10 years ago
Last Released	3 weeks ago



Django

Baptiste Darthenay | 17,031,915 | ★★★★★ (139)

Beautiful syntax and scoped snippets for perfectionists with deadlines

[Disable](#) [Uninstall](#) [Auto Update](#)

[DETAILS](#) [FEATURES](#) [CHANGELOG](#)

Django extension for Visual Studio Code

Beautiful syntax and scoped snippets for perfectionists with deadlines

```
{% load static %}{% meta %}<!DOCTYPE html>
<html lang="en">
<head {% meta_namespaces %}>
  <meta charset="UTF-8">
  <title>{{ title }}</title>
</head>
<body {% if body.dark %}class="dark"{% endif %}>
  {% block content %}
    {# this should be overwritten #}
    <p class="content{% if large %} large{% endif %}">Example paragraph for {{ user }}</p>
  {% include "core/default.html" with object=dummy/custom filter|join:" " %}
```

Installation

Identifier	batisteo.vscode-django
Version	1.15.0
Last Updated	2 minutes ago
Size	798.40KB

Marketplace

Published	8 years ago
Last Released	2 years ago

Categories



Django Template

bibhasdn | 969,035 | ★★★★★ (17)

Django template language support for Visual Studio Code

[Disable](#) [Uninstall](#) [Auto Update](#)

[DETAILS](#) [FEATURES](#)

Django Templates for Visual Studio Code

This extension adds language colorization support and user snippets for the Django template language to VS Code.

This is based on the [Jinja extension](#) by the awesome [wholroyd](#).

```
{% extends "layout.html" %}
{% block body %}
  <ul>
    {% for user in users %}
      <li><a href="{% url 'user-profile' user.username %}">{{ user.username }}</a></li>
    {% endfor %}
  </ul>
{% endblock body %}
```

Installation

Identifier	bibhasdn.django-html
Version	1.3.0
Last Updated	2 minutes ago
Size	44.74KB

Marketplace

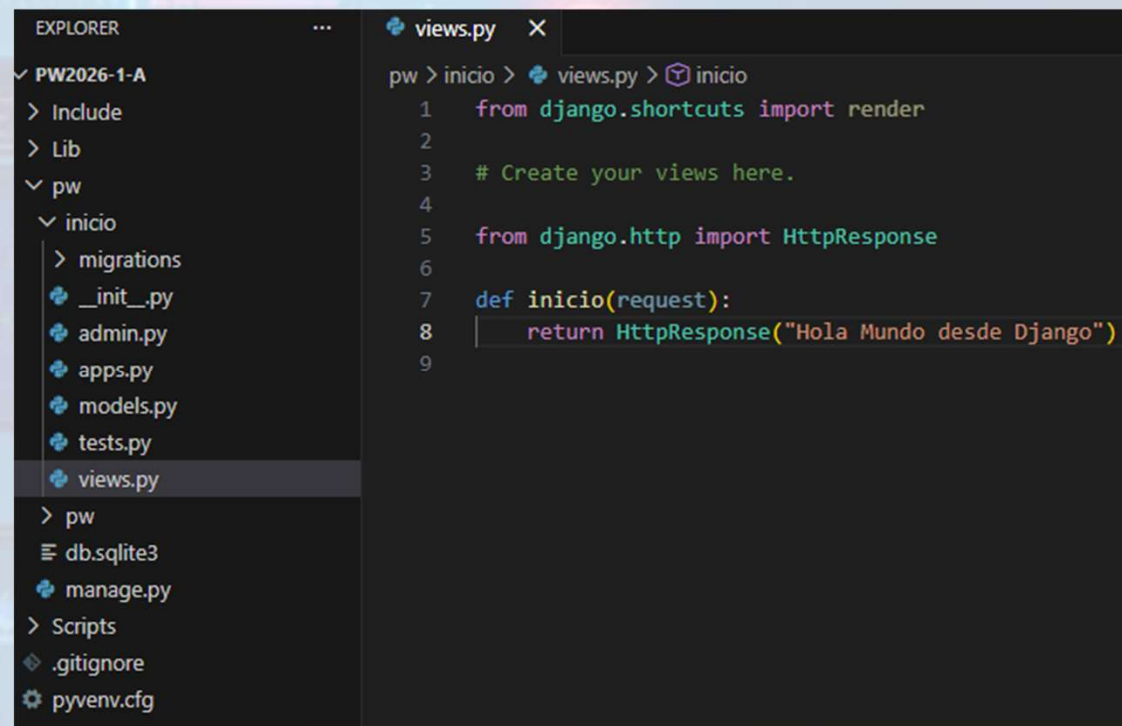
Published	10 years ago
Last Released	7 years ago

Categories

-- ISC Rub

4.1 Introducción al Lenguaje

- Django
 - Vistas
 - Funciones que toman solicitudes HTTP y devuelven respuestas HTTP
 - Pueden haber varias vistas con diferentes propósitos
 - *view.py*



The screenshot shows a code editor with two panels. The left panel, titled 'EXPLORER', displays the project structure for 'PW2026-1-A'. It includes folders like 'Include', 'Lib', and 'pw', and a file explorer for 'inicio' showing 'migrations', '__init__.py', 'admin.py', 'apps.py', 'models.py', 'tests.py', and 'views.py'. The right panel, titled 'views.py', shows the code for the 'inicio' view. It imports 'render' from 'django.shortcuts' and 'HttpResponse' from 'django.http'. The 'inicio' function takes a 'request' and returns an 'HttpResponse' with the text 'Hola Mundo desde Django'.

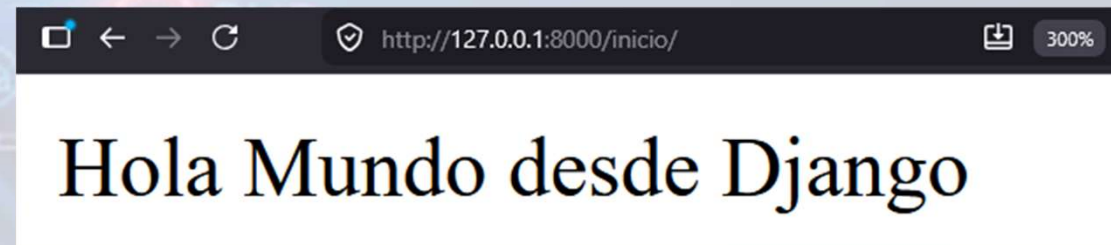
```
EXPLORER
PW2026-1-A
> Include
> Lib
✓ pw
  ✓ inicio
    > migrations
    __init__.py
    admin.py
    apps.py
    models.py
    tests.py
    views.py
  > pw
  db.sqlite3
  manage.py
  Scripts
  .gitignore
  pyenv.cfg

views.py
pw > inicio > views.py > inicio
1  from django.shortcuts import render
2
3  # Create your views here.
4
5  from django.http import HttpResponse
6
7  def inicio(request):
8      return HttpResponse("Hola Mundo desde Django")
9
```

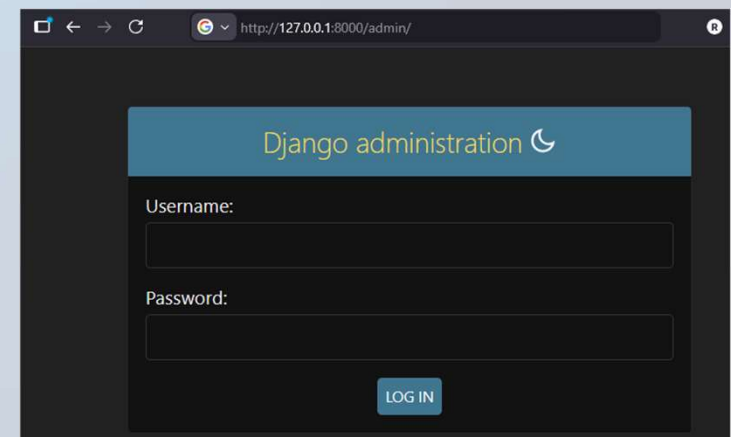
4.1 Introducción al Lenguaje

- Django
 - URLs

```
pw > inicio > urls.py > ...  
1  from django.urls import path  
2  from . import views  
3  
4  urlpatterns = [  
5      path('inicio/', views.inicio, name='inicio'),  
6  ]
```



```
pw > pw > urls.py > ...  
2  URL configuration for pw project.  
3  
4  The `urlpatterns` list routes URLs to views. For more information please see:  
5  https://docs.djangoproject.com/en/6.0/topics/http/urls/  
6  Examples:  
7  Function views  
8      1. Add an import: from my_app import views  
9      2. Add a URL to urlpatterns: path('', views.home, name='home')  
10 Class-based views  
11      1. Add an import: from other_app.views import Home  
12      2. Add a URL to urlpatterns: path('', Home.as_view(), name='home')  
13 Including another URLconf  
14      1. Import the include() function: from django.urls import include, path  
15      2. Add a URL to urlpatterns: path('blog/', include('blog.urls'))  
16 """  
17 from django.contrib import admin  
18 from django.urls import path, include  
19  
20 urlpatterns = [  
21     path('', include('inicio.urls')),  
22     path('admin/', admin.site.urls),  
23 ]  
24
```



4.1 Introducción al Lenguaje

- Django
 - Plantillas
 - Crear directorio *templates* dentro de la app
 - Crear html
 - Modificar *view* de la app
 - Agregar app a lista de apps instaladas
 - Aplicar migración *python manage.py migrate*

```
pw > inicio > views.py > ...
1 from django.shortcuts import render
2
3 # Create your views here.
4
5 from django.http import HttpResponse
6 from django.template import loader
7
8 def inicio(request):
9     #return HttpResponse("Hola Mundo desde Django")
10    template = loader.get_template('plantilla.html')
11    return HttpResponse(template.render())
12
```

EXPLORER

pw2026-1-A

- > Include
- > Lib
- pw
 - inicio
 - > __pycache__
 - > migrations
 - templates
 - plantilla.html
 - __init__.py
 - admin.py
 - apps.py
 - models.py
 - tests.py
 - urls.py
 - views.py
- pw
- db.sqlite3
- manage.py
- > Scripts
- .gitignore
- pyenv.cfg

pw > inicio > templates > plantilla.html

```
1 <!DOCTYPE html>
2 <html>
3 <body>
4
5     <h1>Hello World!</h1>
6
7     <p>Welcome to my first Django project!</p>
8 </body>
9 </html>
```

```
pw > pw > settings.py > ...
22 # SECURITY WARNING: keep the secret key used in production secret!
23 SECRET_KEY = 'django-insecure-u9inyw&nr&gnu1n*x30zjp5emexdx(0y7z9a==_5e=0ulygaq4'
24
25 # SECURITY WARNING: don't run with debug turned on in production!
26 DEBUG = True
27
28 ALLOWED_HOSTS = []
29
30
31 # Application definition
32
33 INSTALLED_APPS = [
34     'django.contrib.admin',
35     'django.contrib.auth',
36     'django.contrib.contenttypes',
37     'django.contrib.sessions',
38     'django.contrib.messages',
39     'django.contrib.staticfiles',
40     'inicio',
41 ]
```

```
(pw2026-1-A) W:\TEC\2026 ENE-JUN\PROGRAMACION WEB\EJEMPLOS\pw2026-1-A\pw>python manage.py migrate
Operations to perform:
Apply all migrations: admin, auth, contenttypes, sessions
Running migrations:
Applying contenttypes.0001_initial... OK
Applying auth.0001_initial... OK
Applying admin.0001_initial... OK
Applying admin.0002_logentry_remove_auto_add... OK
Applying admin.0003_logentry_add_action_flag_choices... OK
Applying contenttypes.0002_remove_content_type_name... OK
Applying auth.0002_alter_permission_name_max_length... OK
Applying auth.0003_alter_user_email_max_length... OK
Applying auth.0004_alter_user_username_opts... OK
Applying auth.0005_alter_user_last_login_null... OK
Applying auth.0006_require_contenttypes_0002... OK
Applying auth.0007_alter_validators_add_error_messages... OK
Applying auth.0008_alter_user_username_max_length... OK
Applying auth.0009_alter_user_last_name_max_length... OK
Applying auth.0010_alter_group_name_max_length... OK
Applying auth.0011_update_proxy_permissions... OK
Applying auth.0012_alter_user_first_name_max_length... OK
Applying sessions.0001_initial... OK
```

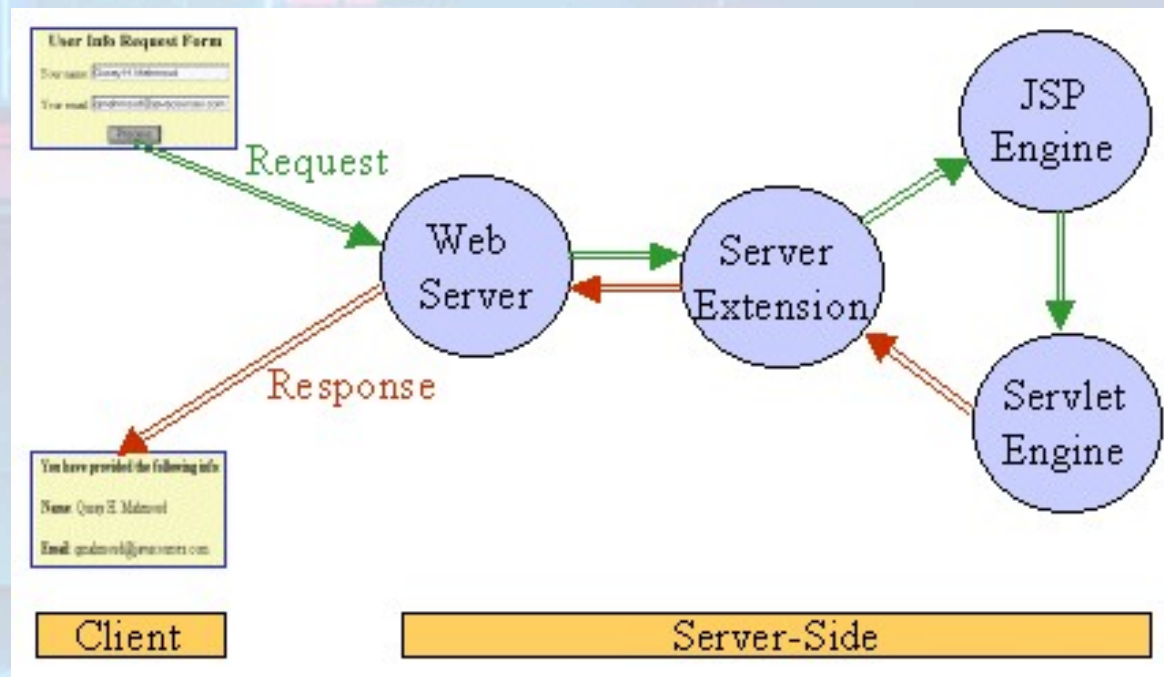
http://127.0.0.1:8000/inicio/ 170%

Hello World!

Welcome to my first Django project!

4.1 Introducción al Lenguaje

- JSP
 - Servlet: Java con HTML
 - JSP: HTML con Java



4.1 Introducción al Lenguaje

- JSP
 - JSP: HTML con Java

com.mysql.cj.jdbc.exceptions.CommunicationsException:
Communications link failure The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the server.

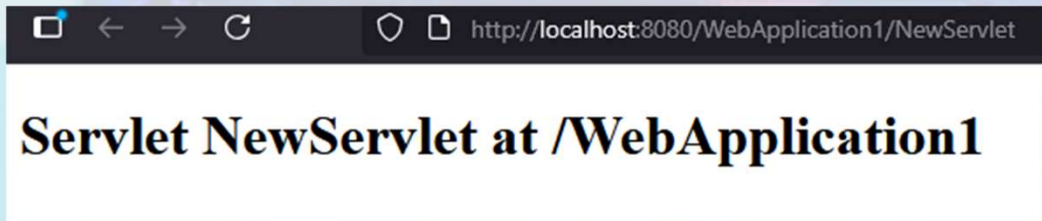
```
<%@page
    contentType = "text/html"
    pageEncoding = "ISO-8859-1"
    session = "true"
    import = "java.sql.*"
    import = "conexionBD.*"
%>

<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>Probar Conexion BD</title>
    </head>
    <body>
        <div>JSP</div>
        <%
            Connection conexion = null; //Variable de tipo Conexion
            ResultSet resultados = null; //Variable para resultados de consultas

            try{
                conexion = conexionBD.abrirConexion(); //Crear conexion de la clase conexionBD
                DatabaseMetaData metaDB = conexion.getMetaData(); //Crear objeto de tipo metadatos con la conexion actual
                out.println("<h1>Conexion Exitosa!</h1>");
                resultados = metaDB.getTables(null, null, "%", new String[]{"TABLE"}); //Obtener metadatos de las tablas de BD
                while(resultados.next()){ //Mientras se tengan resultados en el cursor
                    out.println(resultados.getString("TABLE_NAME") + "<br>"); //Obtener nombre de la tabla
                }
            }
            catch(SQLException e){
                out.println("<h1>" + e + "</h1>");
            }
        %>
    </body>
</html>
```

4.1 Introducción al Lenguaje

- JSP
 - Servlet: Java con HTML



```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

/**
 *
 * @author ruben
 */
@WebServlet(urlPatterns = {"/NewServlet"})
public class NewServlet extends HttpServlet {

    /**
     * Processes requests for both HTTP GET and POST
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet NewServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet NewServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }
}
```

HttpServlet methods. Click on the + sign on the left to edit the code.

Actividad 1

- Considerar la Actividad 1 de la Unidad 3
- Crear una Pagina Web que muestre servicios de un Ingeniero en Sistemas Computacionales (Sin Formulario)
- Opciones de desarrollo:
 1. Usar Django si no lo has implementado
 2. Usar JSP si no lo has implementado
- Notas:
 - Solo usar JavaScript, HTML y CSS (Sin Librerías ni Frameworks)
 - Separar código fuente en un archivo html, en .js y .css
 - Usar *document.createElement()*, *element.appendChild()*
 - Validar HTML

4.2 Estructuras de Control

- Django
 - Variables

```
from django.shortcuts import render
import datetime

# Create your views here.

from django.http import HttpResponse
from django.template import loader

def inicio(request):
    template = loader.get_template('inicio.html')
    return HttpResponse(template.render())

def test(request):
    ahora = datetime.datetime.now()
    contexto = {
        'hora_servidor': ahora,
    }
    return render(request, 'test.html', contexto)
```

```
<!DOCTYPE html>
<html>
<body>

<h1>TEST</h1>
<p>Welcome to my first Django project!</p>

<p>
    La hora actual del servidor es: <strong>{{ hora_servidor | date:"H:i:s" }}</strong>
</p>

{% with varTemplate="Variable en Plantilla" %}
<h1>Esto es una: {{ varTemplate }}</h1>
{% endwith %}

<a href="{% url 'inicio' %}">Ir a INICIO</a>

</body>
</html>
```



4.2 Estructuras de Control

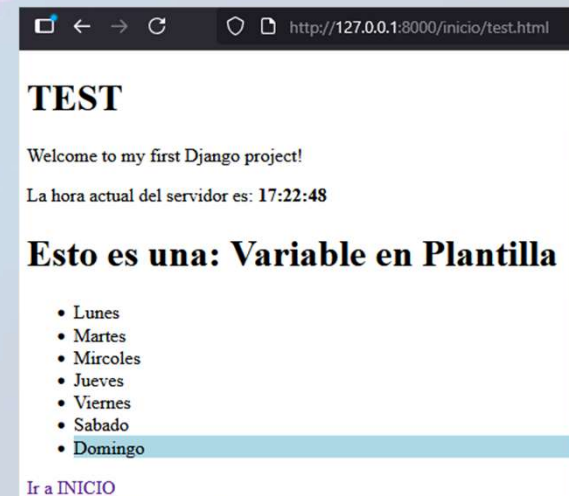
- Django
 - Etiquetas

Tag	Description
autoescape	Specifies if autoescape mode is on or off
block	Specifies a block section
comment	Specifies a comment section
csrf_token	Protects forms from Cross Site Request Forgeries
cycle	Specifies content to use in each cycle of a loop
debug	Specifies debugging information
extends	Specifies a parent template
filter	Filters content before returning it
firstof	Returns the first not empty variable
for	Specifies a for loop
if	Specifies a if statement
ifchanged	Used in for loops. Outputs a block only if a value has changed since the last iteration
include	Specifies included content/template
load	Loads template tags from another library
lorem	Outputs random text
now	Outputs the current date/time
regroup	Sorts an object by a group
resetcycle	Used in cycles. Resets the cycle
spaceless	Removes whitespace between HTML tags
templatetag	Outputs a specified template tag
url	Returns the absolute URL part of a URL
verbatim	Specifies contents that should not be rendered by the template engine
widthratio	Calculates a width value based on the ratio between a given value and a max value
with	Specifies a variable to use in the block

4.2 Estructuras de Control

- Django
 - Ciclos
 - *forloop*
 - *forloop.counter*
 - *forloop.counter0*
 - *forloop.first*
 - *forloop.last*
 - *forloop.parentloop*
 - *forloop.revcounter*
 - *forloop.revcounter0*

```
<ul>
  {% for x in dias %}
    <li>
      {% if forloop.last %}
        style='background-color:lightblue;'
      {% endif %}
    >{{ x }}</li>
  {% endfor %}
</ul>
```



4.2 Estructuras de Control

- JSP

```
if(login.equals("registro@cyes.mx") && pass.equals("3a46f86646ab920214392cd55feac511")){  
    out.println("<script type='text/JavaScript' language='JavaScript' charset='ISO-8859-1'>");  
    out.println("cambiarPagina('Persona/registro.jsp');");  
    out.println("</script>");  
}  
else{  
    try {  
        s = new Sesion();  
        declaracion = con.abrirBase(s);  
        comunicacion = true;  
    }  
    catch (Exception b) {  
        error e = new error();  
        out.println(e.imprimirError(b, "1", ""));  
    }  
    if(comunicacion){
```

4.2 Estructuras de Control

- JSP

```
public Statement abrirBase(String nombre_usuario, String password) throws ClassNotFoundException, SQLException{
    Class.forName(driver());
    conexion = DriverManager.getConnection(server() + dataBase(), nombre_usuario, password);
    declaracion = conexion.createStatement();
    return declaracion;
}

public Statement abrirBase(Sesion s) throws ClassNotFoundException, SQLException{
    Class.forName(driver());
    conexion = DriverManager.getConnection(server() + dataBase(), s.nombre_usuario, s.pass);
    declaracion = conexion.createStatement();
    return declaracion;
}
```

4.2 Estructuras de Control

- JSP

```
HttpSession session = request.getSession(true);
session.setMaxInactiveInterval(7200);
session.setAttribute("login", login2);
session.setAttribute("tipo_usuario", tipo_usuario);
session.setAttribute("pass", pass2);
session.setAttribute("aviso", "-1");
s = null;
con = new ConexionCyes();
s = new Sesion(login2, pass2);
declaracion = con.abrirBase(s);
consulta = "BEGIN";
declaracion.executeUpdate(consulta);
consulta = "INSERT INTO acceso_sistema "
        + "(persona_login, ingreso, penultimo_m, ultimo_m, sesion, forma_cierre) "
        + "VALUES ('"+login2+"', "
        + ""+tiempo.fechaHora()+", 0, "+tiempo.fechaHora()+", 1, 'NINGUNA');"
declaracion.executeUpdate(consulta);
consulta = "SELECT last_insert_id();";
resultados = declaracion.executeQuery(consulta);
if(resultados.next())
    session.setAttribute("numero_acceso", resultados.getString(1));
consulta = "COMMIT";
declaracion.executeQuery(consulta);
out.println("<script type='text/JavaScript' language='JavaScript' charset='ISO-8859-1'>");
out.println("cambiarPagina('inicio.jsp');");
out.println("</script>");
```


4.3 Tratamiento de Formularios

- Django

```
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>ADD1</title>
</head>
<body>

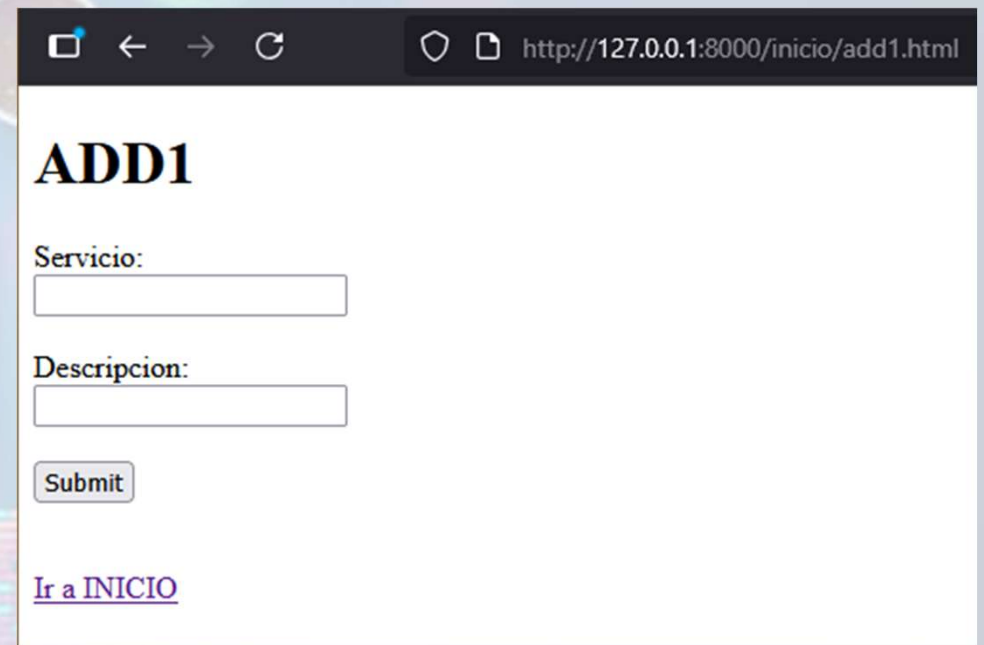
<h1>ADD1</h1>

<form action="addrecord/" method="post">
  {% csrf_token %}
  Servicio:<br>
  <input name="servicio" required>
  <br><br>
  Descripcion:<br>
  <input name="descripcion" required>
  <br><br>
  <input type="submit" value="Submit">
</form>

<br><br>

<a href="{% url 'inicio' %}">Ir a INICIO</a>

</body>
</html>
```



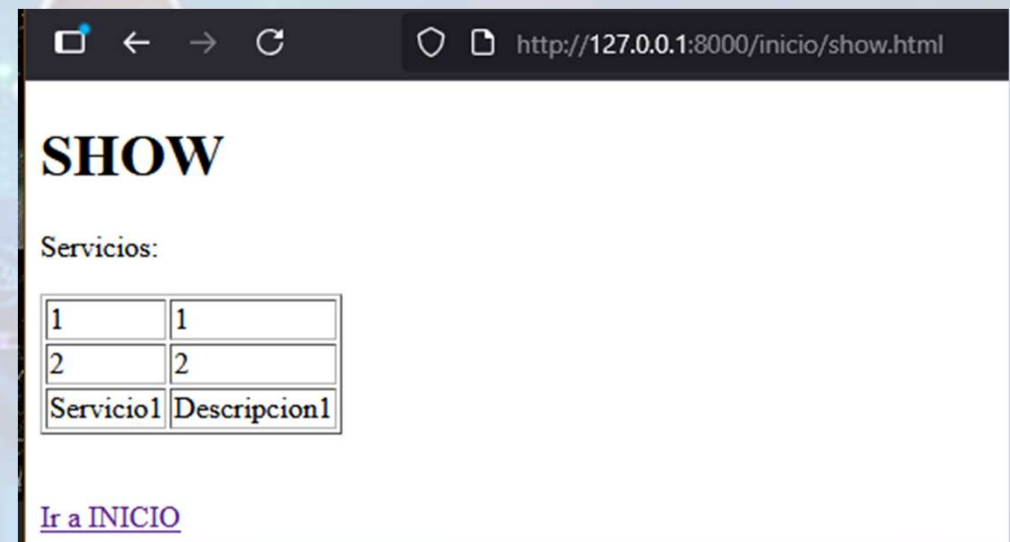
The screenshot shows a web browser window with the address bar displaying `http://127.0.0.1:8000/inicio/add1.html`. The page title is "ADD1". The form contains two input fields: "Servicio:" and "Descripcion:". Below the "Descripcion:" field is a "Submit" button. At the bottom of the form, there is a link labeled "Ir a INICIO".

4.3 Tratamiento de Formularios

- Django

```
def addrecord(request):
    servicio = request.POST['servicio']
    descripcion = request.POST['descripcion']
    lista_actual = request.session.get('servicios', [])
    lista_actual.append({'nombre': servicio, 'descripcion': descripcion})
    request.session['servicios'] = lista_actual
    return HttpResponseRedirect(reverse('show'))
```

```
def show(request):
    servicios_guardados = request.session.get('servicios', [])
    contexto = {
        'lista_servicios': servicios_guardados
    }
    return render(request, 'show.html', contexto)
```



The screenshot shows a web browser window with the address bar displaying `http://127.0.0.1:8000/inicio/show.html`. The page content includes the heading **SHOW**, followed by the text "Servicios:". Below this is a table with two columns. The first column contains the numbers "1" and "2", and the second column contains the numbers "1" and "2". Below the table, there are labels "Servicio1" and "Descripcion1". At the bottom of the page, there is a link labeled "Ir a INICIO".

1	1
2	2
Servicio1	Descripcion1

[Ir a INICIO](#)

4.3 Tratamiento de Formularios

- JSP

```
<form action="registro_bd.jsp" method="post" name="cuenta" id="cuenta">
  <table class="form_tabla">
    <tr>
      <td class="form_izq">
        E-mail:
      </td>
      <td class="form_der">
        <input id="e_mail" name="e_mail" type="text" size="40" maxlength="60" class="form_elemento" onkeydown="JavaScript:validarE_mail()" onkeyup="JavaScript:validarE_mail()" />
        <input id="e_mail_repetido" name="e_mail_repetido" type="text" value="correcto" style="display:none" />
        <span id="salida_e_mail" class="error-message"></span>
      </td>
    </tr>
    <tr>
      <td class="form_izq">
        Nombre:
      </td>
      <td class="form_der">
        <input id="nombre" name="nombre" type="text" size="30" maxlength="40" class="form_elemento" onkeydown="JavaScript:validarNombre()" onkeyup="JavaScript:validarNombre()" />
      </td>
    </tr>
  </table>
</form>
```

4.3 Tratamiento de Formularios

- JSP

```
function validarE_mail(){
    if($("#suscriptor").validate().element("#e_mail")){
        var mail = formatoDireccion(document.getElementById("e_mail").value);
        if( window.XMLHttpRequest )
            ajax = new XMLHttpRequest(); // No Internet Explorer
        else
            ajax = new ActiveXObject("Microsoft.XMLHTTP"); // Internet Explorer
        ajax.onreadystatechange = function respuesta_e_mail(){
            if( ajax.readyState == 4 ){
                if( ajax.status == 200 ){
                    document.getElementById("salida_e_mail").innerHTML = "<b>"+ajax.responseText+"</b>";
                    var error = document.getElementById("e_mail_repetido");
                    if(ajax.responseText.indexOf("repetido") == -1){
                        $("#salida_e_mail").removeClass("error-message").addClass("correct-message");
                        document.getElementById("e_mail_repetido").value = ajax.responseText;
                    }
                    else{
                        $("#salida_e_mail").removeClass("correct-message").addClass("error-message");
                        document.getElementById("e_mail_repetido").value = '';
                    }
                }
            }
        };
        ajax.open( "GET", "validacion_e_mail.jsp?e="+mail+"&z="+Math.random(), true );
        ajax.send( "" );
    }
    else{
        document.getElementById("salida_e_mail").innerHTML = "";
        document.getElementById("e_mail_repetido").value = "correcto";
    }
}
```

```
if(comunicacion){
    try{
        e_mail = request.getParameter("e").toString();
    }
    catch(Exception c){
        error e = new error();
        out.println(e.imprimirError(c, "1", "../"));
    }
    try{
        login2 = request.getParameter("login2").toString();
    }
    catch(Exception d){}
    try{
        consulta = "SELECT * FROM persona WHERE login='"+e_mail+"' "
            + "AND login='"+login2+"'";
        resultados = declaracion.executeQuery(consulta);
        if(resultados.next())
            out.println("E-mail: " + resultados.getString("login") + " válido");
        else{
            consulta = "SELECT * FROM persona WHERE login='"+e_mail+"'";
            resultados = declaracion.executeQuery(consulta);
            if(resultados.next())
                out.println("E-mail: " + resultados.getString("login") + " repetido");
            else
                out.println("E-mail: " + e_mail + " válido");
        }
    }
    catch(Exception f){
        error e = new error();
        out.println(e.imprimirError(f, "1", "../"));
    }
    try{
        con.cerrarBase();
    }
    catch (Exception z) {
        error e = new error();
        out.println(e.imprimirError(z, "1", "../"));
    }
}
```

4.4 Manejo de Objetos del Servidor

- Django
 - *settings.py*

```
"""
Django settings for pw3 project.

Generated by 'django-admin startproject' using Django 5.2.7.

For more information on this file, see
https://docs.djangoproject.com/en/5.2/topics/settings/

For the full list of settings and their values, see
https://docs.djangoproject.com/en/5.2/ref/settings/
"""

from pathlib import Path

# Build paths inside the project like this: BASE_DIR / 'subdir'.
BASE_DIR = Path(__file__).resolve().parent.parent

# Quick-start development settings - unsuitable for production
# See https://docs.djangoproject.com/en/5.2/howto/deployment/checklist/

# SECURITY WARNING: keep the secret key used in production secret!
SECRET_KEY = "django-insecure-e3p0o4)288bnfj-xk6ok=qov2@61t2r)ud72n7(l4l1(ugw7bs"

# SECURITY WARNING: don't run with debug turned on in production!
DEBUG = True

ALLOWED_HOSTS = []

# Application definition

INSTALLED_APPS = [
    "django.contrib.admin",
    "django.contrib.auth",
    "django.contrib.contenttypes",
    "django.contrib.sessions",
    "django.contrib.messages",
    "django.contrib.staticfiles",
    "inicio"
]
```


4.4 Manejo de Objetos del Servidor

- Django
 - *settings.py*

```
MIDDLEWARE = [  
    "django.middleware.security.SecurityMiddleware",  
    "django.contrib.sessions.middleware.SessionMiddleware",  
    "django.middleware.common.CommonMiddleware",  
    "django.middleware.csrf.CsrfViewMiddleware",  
    "django.contrib.auth.middleware.AuthenticationMiddleware",  
    "django.contrib.messages.middleware.MessageMiddleware",  
    "django.middleware.clickjacking.XFrameOptionsMiddleware",  
]  
  
ROOT_URLCONF = "pw3.urls"  
  
TEMPLATES = [  
    {  
        "BACKEND": "django.template.backends.django.DjangoTemplates",  
        "DIRS": [],  
        "APP_DIRS": True,  
        "OPTIONS": {  
            "context_processors": [  
                "django.template.context_processors.request",  
                "django.contrib.auth.context_processors.auth",  
                "django.contrib.messages.context_processors.messages",  
            ],  
        },  
    ],  
]  
  
WSGI_APPLICATION = "pw3.wsgi.application"
```

4.4 Manejo de Objetos del Servidor

- Django
 - *settings.py*

```
# Database
# https://docs.djangoproject.com/en/5.2/ref/settings/#databases

DATABASES = {
    "default": {
        "ENGINE": "django.db.backends.sqlite3",
        "NAME": BASE_DIR / "db.sqlite3",
    }
}

# Password validation
# https://docs.djangoproject.com/en/5.2/ref/settings/#auth-password-validators

AUTH_PASSWORD_VALIDATORS = [
    {
        "NAME": "django.contrib.auth.password_validation.UserAttributeSimilarityValidator",
    },
    {
        "NAME": "django.contrib.auth.password_validation.MinimumLengthValidator",
    },
    {
        "NAME": "django.contrib.auth.password_validation.CommonPasswordValidator",
    },
    {
        "NAME": "django.contrib.auth.password_validation.NumericPasswordValidator",
    },
]

# Internationalization
# https://docs.djangoproject.com/en/5.2/topics/i18n/

LANGUAGE_CODE = "en-us"

TIME_ZONE = "UTC"

USE_I18N = True

USE_TZ = True

# Static files (CSS, JavaScript, Images)
# https://docs.djangoproject.com/en/5.2/howto/static-files/

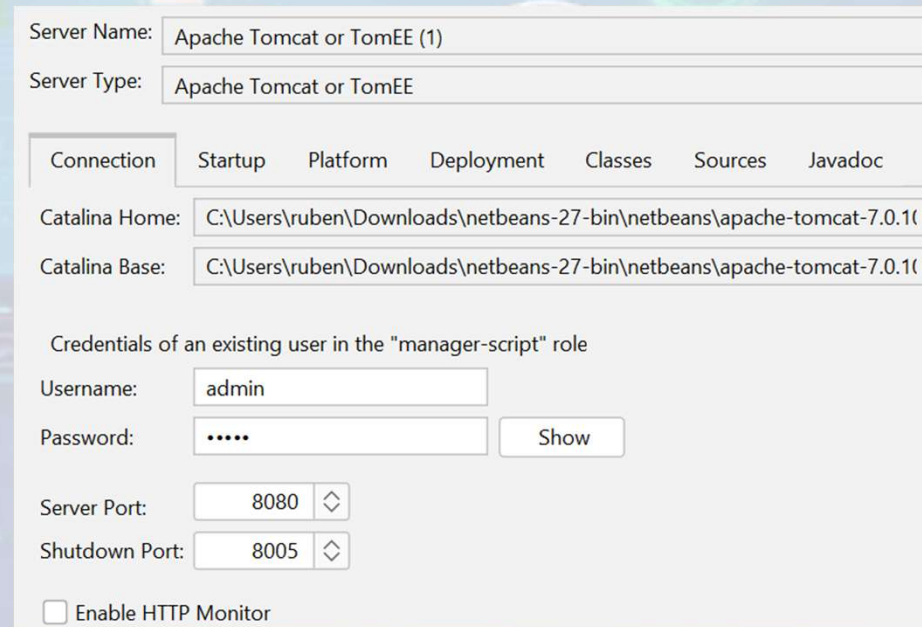
STATIC_URL = "static/"

# Default primary key field type
# https://docs.djangoproject.com/en/5.2/ref/settings/#default-auto-field

DEFAULT_AUTO_FIELD = "django.db.models.BigAutoField"
```

4.4 Manejo de Objetos del Servidor

- JSP



The screenshot shows the 'New Project' dialog in NetBeans, specifically the 'Server' tab. The configuration is as follows:

Connection	Startup	Platform	Deployment	Classes	Sources	Javadoc
Catalina Home: C:\Users\ruben\Downloads\netbeans-27-bin\netbeans\apache-tomcat-7.0.100						
Catalina Base: C:\Users\ruben\Downloads\netbeans-27-bin\netbeans\apache-tomcat-7.0.100						
Credentials of an existing user in the "manager-script" role						
Username: admin						
Password:		Show				
Server Port: 8080		⬇				
Shutdown Port: 8005		⬇				
☐ Enable HTTP Monitor						

4.4 Manejo de Objetos del Servidor

- JSP
- *server.xml*

```
<?xml version='1.0' encoding='utf-8'?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<!-- Note: A "Server" is not itself a "Container", so you may not
define subcomponents such as "Valves" at this level.
Documentation at /docs/config/server.html
-->
<Server port="8005" shutdown="SHUTDOWN">
  <Listener className="org.apache.catalina.startup.VersionLoggerListener" />
  <!-- Security listener. Documentation at /docs/config/listeners.html
  <Listener className="org.apache.catalina.security.SecurityListener" />
  <!-- APR library loader. Documentation at /docs/apr.html -->
  <Listener className="org.apache.catalina.core.AprLifecycleListener" SSLEngine="on" />
  <!-- Initialize Jasper prior to webapps are loaded. Documentation at /docs/jasper-howto.html -->
  <Listener className="org.apache.catalina.core.JasperListener" />
  <!-- Prevent memory leaks due to use of particular java/javax APIs-->
  <Listener className="org.apache.catalina.core.JreMemoryLeakPreventionListener" />
  <Listener className="org.apache.catalina.mbeans.GlobalResourcesLifecycleListener" />
  <Listener className="org.apache.catalina.core.ThreadLocalLeakPreventionListener" />

  <!-- Global JNDI resources
  Documentation at /docs/jndi-resources-howto.html
  -->
  <GlobalNamingResources>
    <!-- Editable user database that can also be used by
    UserDatabaseRealm to authenticate users
    -->
    <Resource name="UserDatabase" auth="Container"
      type="org.apache.catalina.UserDatabase"
      description="User database that can be updated and saved"
      factory="org.apache.catalina.users.MemoryUserDatabaseFactory"
      pathname="conf/tomcat-users.xml" />
  </GlobalNamingResources>
```

```
<Service name="Catalina">

  <!--The connectors can use a shared executor, you can define one or more named thread pools-->
  <!--
  <Executor name="tomcatThreadPool" namePrefix="catalina-exec-"
    maxThreads="150" minSpareThreads="4"/>
  -->

  <!-- A "Connector" represents an endpoint by which requests are received
  and responses are returned. Documentation at :
  Java HTTP Connector: /docs/config/http.html (blocking & non-blocking)
  Java AJP Connector: /docs/config/ajp.html
  APR (HTTP/AJP) Connector: /docs/apr.html
  Define a non-SSL HTTP/1.1 Connector on port 8080
  -->
  <Connector port="8080" protocol="HTTP/1.1"
    connectionTimeout="20000"
    redirectPort="8443" />

  <!-- A "Connector" using the shared thread pool-->
  <!--
  <Connector executor="tomcatThreadPool"
    port="8080" protocol="HTTP/1.1"
    connectionTimeout="20000"
    redirectPort="8443" />
  -->

  <!-- Define an SSL HTTP/1.1 Connector on port 8443
  This connector uses the BIO implementation that requires the JSSE
  style configuration. When using the APR/native implementation, the
  OpenSSL style configuration is required as described in the APR/native
  documentation -->
  <!--
  <Connector port="8443" protocol="org.apache.coyote.http11.Http11Protocol"
    maxThreads="150" SSLEnabled="true" scheme="https" secure="true"
    clientAuth="false" sslProtocol="TLS" />
  -->

  <!-- Define an AJP 1.3 Connector on port 8009 -->
  <!--
  <Connector protocol="AJP/1.3"
    address=":::1"
    port="8009"
    redirectPort="8443" />
  -->
```


Actividad 2

- Considerar la Actividad anterior
- En la nueva pagina, crear un formulario con los elementos necesarios para agregar un nuevo producto
- Aplicar todas las validaciones y casos especiales para poder almacenar el nuevo producto
- Si hay errores en el formulario, proponer una alternativa para poder visualizar el mensaje de error y hacer sugerencia para la corrección
- Regresar al *index.html* y ver reflejado el nuevo objeto
- Opciones de desarrollo:
 1. Usar Django si no lo has implementado
 2. Usar JSP si no lo has implementado
- Notas:
 - Usar almacenamiento local o en sesión dependiendo el lenguaje del servidor
 - Validar HTML

4.5 Creación de Clases

- Django
 - Modelos
 - Instalar cliente BD: *pip install mysqlclient*
 - Modificar *settings.py*

```
# Database
# https://docs.djangoproject.com/en/5.2/ref/settings/#databases

DATABASES = {
    "default": {
        "ENGINE": "django.db.backends.mysql",
        "NAME": "pw",
        "USER": "admin",
        "PASSWORD": "admin",
        "HOST": "localhost",
        "PORT": "3306"
    }
}
```

4.5 Creación de Clases

- Django
 - Modelos
 - Crear Modelo
 - Crear Tabla desde Django
 - Aplicar Migración

```
from django.db import models

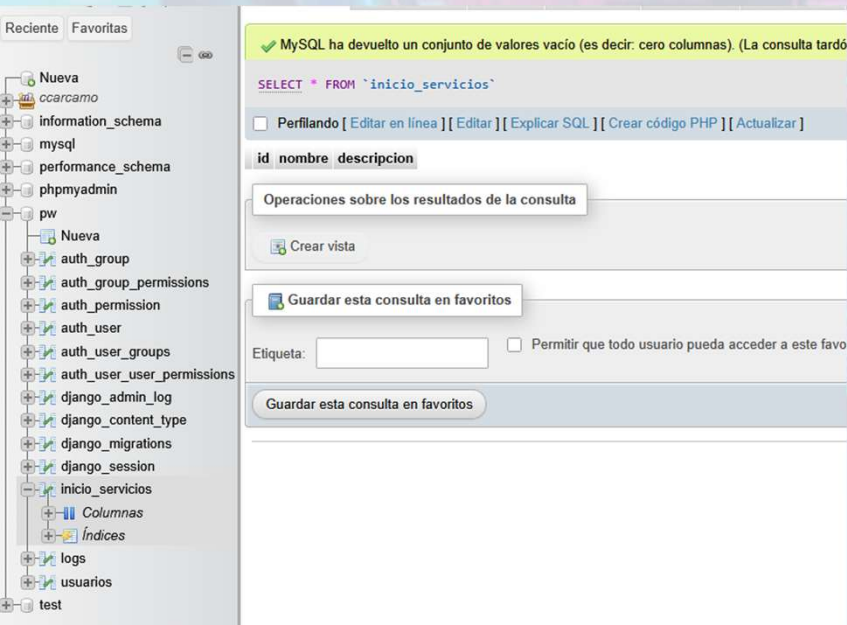
# Create your models here.

class Servicios(models.Model):
    nombre = models.CharField(max_length=200)
    descripcion = models.CharField(max_length=200)
```

```
(pw2025-03) W:\TEC\2025 AGO-DIC\PROGRAMACION WEB\EJEMPLOS\pw2025-03\pw3>python manage.py makemigrations
Migrations for 'inicio':
  inicio\migrations\0001_initial.py
    - Create model Servicios
```

```
(pw2025-03) W:\TEC\2025 AGO-DIC\PROGRAMACION WEB\EJEMPLOS\pw2025-03\pw3>python manage.py migrate
System check identified some issues:
```

```
WARNINGS:
?: (mysql.W002) MariaDB Strict Mode is not set for database connection 'default'
  HINT: MariaDB's Strict Mode fixes many data integrity problems in MariaDB, such as data truncation upon insertion, by escalating warnings into errors. It is strongly recommended you activate it. See
: https://docs.djangoproject.com/en/4.1/ref/databases/#mysql-sql-mode
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, inicio, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying inicio.0001_initial... OK
  Applying sessions.0001_initial... OK
```



4.5 Creación de Clases

- Django
 - Modelos
 - Crear Formulario

```
from django import forms
from .models import Servicios

class ServicioFormulario(forms.ModelForm):
    class Meta:
        model = Servicios
        fields = ['nombre', 'descripcion']
```

```
<!DOCTYPE html>
<html lang="es">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>AGREGAR SERVICIO BD</title>
</head>

<body>

    <h1>AGREGAR NUEVO SERVICIO BD</h1>

    <form method="post">
        {% csrf_token %}

        {{ form.as_p }}

        <input type="submit" value="GUARDAR SERVICIO BD">
    </form>

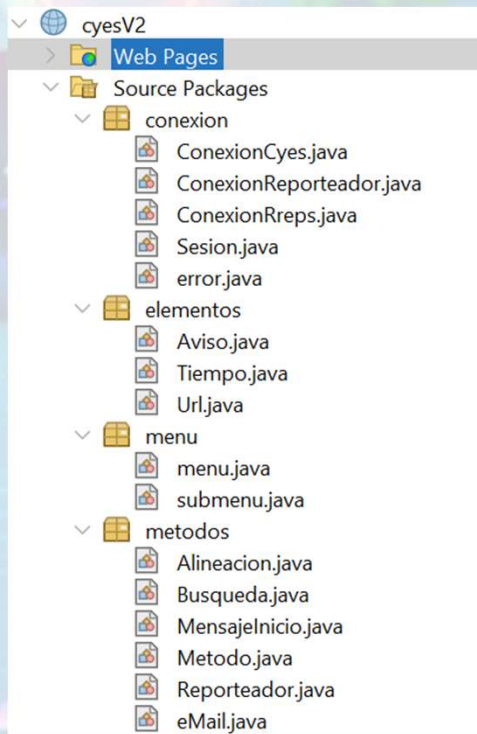
    <br><br>

    <a href="{% url 'inicio' %}">IR A INICIO</a>

</body>
</html>
```

4.5 Creación de Clases

- JSP



4.5 Creación de Clases

- JSP

```

public class error {

    private String html_error;

    public error(){
        html_error = "";
    }

    public String imprimirError(Exception e, String atras, String directorio){
        html_error = html_error +
            "\t<table class='error_tabla'>" +
            "\t\t<tr>" +
            "\t\t\t<td>" +
            "\t\t\t\t<br />" +
            "\t\t\t\t\t" + e +
            "\t\t\t\t\t</td>" +
            "\t\t</tr>" +
            "\t\t<tr>" +
            "\t\t\t<td>" +
            "\t\t\t\t\t<table class='error_icono_tabla' align='center'>" +
            "\t\t\t\t\t\t<tr>" +
            "\t\t\t\t\t\t\t<td class='error_icono'>" +
            "\t\t\t\t\t\t\t\t<a href='javascript:history.go(\"-"+atras+")'>" +
            "\t\t\t\t\t\t\t\t\t<img src='"+directorio+"Imagenes/Iconos/4_5.png' alt='Atras' />" +
            "\t\t\t\t\t\t\t\t\t<br />" +
            "\t\t\t\t\t\t\t\t\t\tAtras" +
            "\t\t\t\t\t\t\t\t\t</a>" +
            "\t\t\t\t\t\t\t\t</td>" +
            "\t\t\t\t\t\t\t\t\t<td class='error_icono'>" +
            "\t\t\t\t\t\t\t\t\t\t<a href='"+directorio+"inicio.jsp'>" +
            "\t\t\t\t\t\t\t\t\t\t\t<img src='"+directorio+"Imagenes/Iconos/9.png' alt='Inicio' />" +
            "\t\t\t\t\t\t\t\t\t\t\t<br />" +
            "\t\t\t\t\t\t\t\t\t\t\t\tInicio" +
            "\t\t\t\t\t\t\t\t\t\t\t</a>" +
            "\t\t\t\t\t\t\t\t\t\t\t</td>" +
            "\t\t\t\t\t\t\t\t</tr>" +
            "\t\t\t\t\t\t\t</table>" +
            "\t\t\t\t\t</td>" +
            "\t\t\t\t\t</tr>" +
            "\t\t\t\t\t</table>";
        return html_error;
    }
}

```


4.6 Acceso a Datos

- Django

```
def showBD(request):
    todos_los_servicios = Servicios.objects.all()

    contexto = {
        'lista_servicios': todos_los_servicios
    }
    return render(request, 'showBD.html', contexto)
```

```
<!DOCTYPE html>
<html lang="es">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>MOSTRAR SERVICIO BD</title>
</head>

<body>

    <h1>SHOW</h1>

    <p>SERVICIOS BD:</p>

    <table border="1">
        {% for servicio in lista_servicios %}
            <tr>
                <td>{{ servicio.nombre }}</td>
                <td>{{ servicio.descripcion }}</td>
            </tr>
        {% endfor %}
    </table>

    <br><br>

    <a href="{% url 'inicio' %}">IR A INICIO</a>

</body>
</html>
```

4.6 Acceso a Datos

- JSP

```
try{
    consulta = "SELECT DISTINCT(aaaa.c_nombre) AS nombre, aaaa.suc_carpeta, "
        + "aaaa.c_clave AS clave, aaaa.nombre_reporteador, aaaa.est_clave, "
        + "aaaa.est_nombre, aaaa.empresa_clave FROM (SELECT b.suc_clave AS clave, b.suc_nombre AS nombre, "
        + "b.suc_carpeta, b.c_clave, b.c_nombre, b.nombre_reporteador, b.est_clave, "
        + "e.nombre AS est_nombre, b.empresa_clave FROM estado e INNER JOIN(SELECT a.suc_clave, a.suc_nombre, "
        + "a.suc_carpeta, a.c_clave, c.nombre AS c_nombre, c.nombre_reporteador, "
        + "c.estado_clave AS est_clave, c.empresa_clave FROM ciudad c INNER JOIN (SELECT s.clave AS suc_clave, "
        + "s.nombre AS suc_nombre, s.carpeta AS suc_carpeta, s.ciudad_clave AS c_clave "
        + "FROM persona_atiende_sucursal pas INNER JOIN sucursal s "
        + "ON pas.sucursal_clave=s.clave AND pas.persona_login='"+login+"' AND s.carpeta!='Fact') a ON a.c_clave=c.clave";

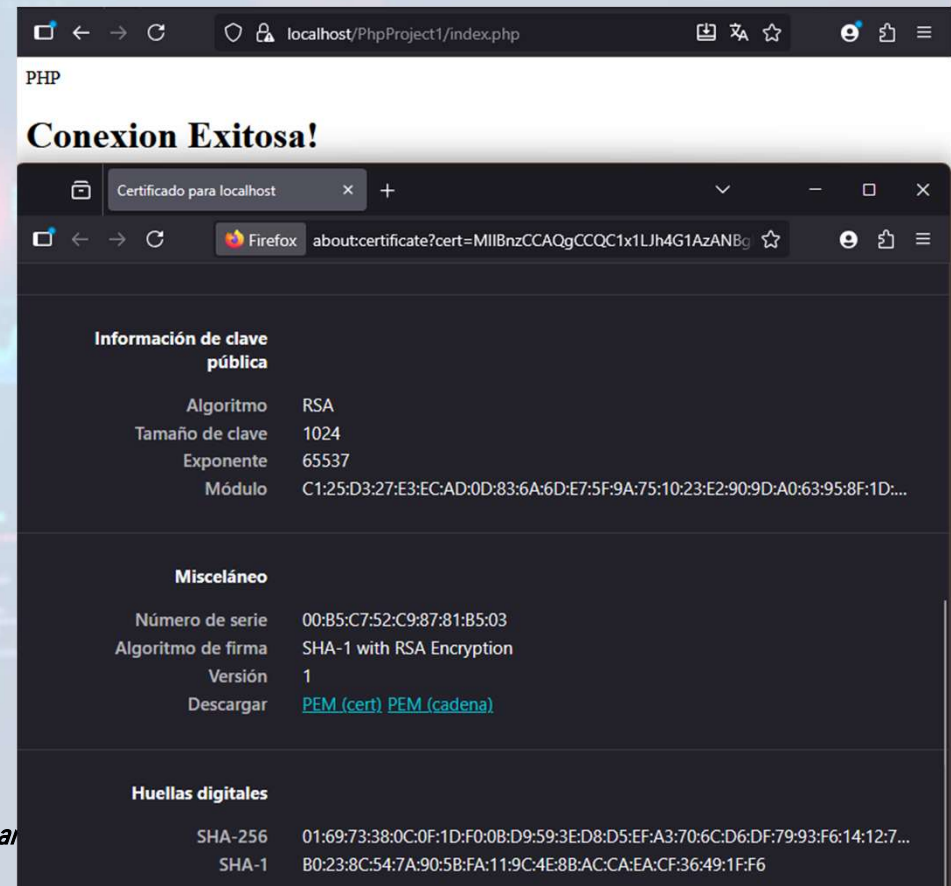
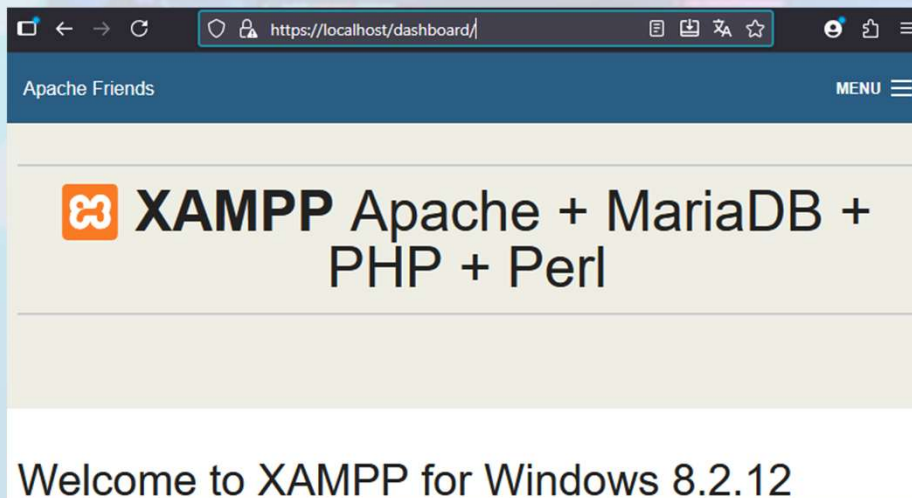
    if(!empresa.equals(""))
        consulta = consulta + " AND c.empresa_clave="+empresa;
    if(!estado2.equals(""))
        consulta = consulta + " AND c.estado_clave="+estado2;
    consulta = consulta + ") b ON b.est_clave=e.clave) aaaa";
    consulta = "SELECT * FROM (" + consulta + ") e";
    consulta = elementos.consultaBusqueda(consulta, declaracion, "nombre");
    resultados = declaracion.executeQuery(consulta);
    if(resultados.next()){
        ConexionReps reporteador_mysql = new ConexionReps("");
        Statement declaracion_rreps = reporteador_mysql.abrirBase();
        ResultSet resultados_rreps = null;
        String consulta_sql_rreps = "";
        resultados.previous();
        out.println("<div id='sucursales_reporteador' style='overflow: auto; height: 406px; vertical-align: top;'>");
        out.println("<table class='lista_elementos'>");
        while(resultados.next()){
            resultados.previous();
            out.println("<tr>");
            for(int i=0; i<3; i++){
                if(resultados.next()){
                    out.println("<td>");
                    out.println("<table class='lista_elemento'>");
                    out.println("<tr>");
                    out.println("<td rowspan='4' class='lista_elemento_img'>");
                    out.println("<div id='ciudad'+resultados.getString('clave')+'>");
                    out.println("<img src='../Imagenes/Iconos/ReportOff.png' alt='"+resultados.getString('nombre')";
                    out.println("</div>");
                    out.println("</td>");
                    out.println("<td class='lista_elemento_texto'>");
```

Actividad 3

- De la actividad anterior, almacenar los servicios en una base de datos diferente a SQLite
- Replicar funcionalidad
- Opciones de desarrollo aplicando modelos y clases:
 1. Usar Django si no lo has implementado
 2. Usar JSP si no lo has implementado
- Crear una nota con las diferencias entre la diferencia de programación del lado del cliente, del servidor, almacenamiento local y con orígenes de datos
- Tomar referencias de estas 3 actividades para completar la entrega de la Practica de la Unidad 4
- Notas:
 - Validar HTML

Conexión Segura

- Apache
 - Viene directo, solo agregar *https* al inicio
 - <https://localhost/dashboard/>
 - <https://localhost/phpmyadmin/>



Conexión Segura

- Django

- Instalar librería *django-sslserver*

- Versión Python: 3.11.14

- Versión Django: 4.1.13

- Añadir librería en *INSTALLED_APPS* dentro del archivo *settings.py*

- Iniciar servidor SSL

```
INSTALLED_APPS = [  
    "django.contrib.admin",  
    "django.contrib.auth",  
    "django.contrib.contenttypes",  
    "django.contrib.sessions",  
    "django.contrib.messages",  
    "django.contrib.staticfiles",  
    "inicio",  
    "sslserver"  
]
```

```
(pw2025-03) W:\TEC\2025 AGO-DIC\PROGRAMACION WEB\EJEMPLOS\pw2025-03\pw3>pip install django-sslserver  
Collecting django-sslserver  
  Downloading django_sslserver-0.22-py3-none-any.whl.metadata (1.1 kB)  
Requirement already satisfied: Django>=1.8 in v:\py\anaconda\envs\pw2025-03\lib\site-packages (from  
django-sslserver) (4.1.13)  
Requirement already satisfied: asgiref<4,>=3.5.2 in v:\py\anaconda\envs\pw2025-03\lib\site-packages  
(from Django>=1.8->django-sslserver) (3.10.0)  
Requirement already satisfied: sqlparse>=0.2.2 in v:\py\anaconda\envs\pw2025-03\lib\site-packages (f  
rom Django>=1.8->django-sslserver) (0.5.3)  
Requirement already satisfied: tzdata in v:\py\anaconda\envs\pw2025-03\lib\site-packages (from Djang  
o>=1.8->django-sslserver) (2025.2)  
Downloading django_sslserver-0.22-py3-none-any.whl (10 kB)  
Installing collected packages: django-sslserver  
Successfully installed django-sslserver-0.22
```

```
(pw2025-03) W:\TEC\2025 AGO-DIC\PROGRAMACION WEB\EJEMPLOS\pw2025-03\pw3>python manage.py runsslserver  
Watching for file changes with StatReloader  
Validating models...  
  
System check identified no issues (0 silenced).  
November 04, 2025 - 10:52:48  
Django version 4.1.13, using settings 'pw3.settings'  
Starting development server at https://127.0.0.1:8000/  
Using SSL certificate: V:\PY\ANACONDA\envs\pw2025-03\Lib\site-packages\sslserver\certs\development.crt  
Using SSL key: V:\PY\ANACONDA\envs\pw2025-03\Lib\site-packages\sslserver\certs\development.key  
Quit the server with CTRL-BREAK.
```

Firefox about:certificate?cert=MIIGSTCCBDGgAwIBAgIJAPc4Jn

Certificado

localhost

Nombre del interesado	
País	US
Estado/Provincia	DEVELOPMENT STATE
Localidad	DEVELOPMENT CITY
Organización	DEVELOPMENT COMPANY
Unidad organizacional	DJANGO DEVELOPERS
Nombre común	localhost
Dirección de correo electrónico	development@example.com

én Lara

Conexión Segura

- JSP (Tomcat)
 - Crear Certificado
 - Usar *keytool*
 - Responder asistente
 - Guardar *keystore password*
 - Configurar Tomcat *server.xml*

```
C:\Program Files\Java\jdk-17\bin>keytool -genkeypair -alias tomcat7 -keyalg RSA -keystore "C:\Users\ruben\Downloads\netbeans-27-bin\netbeans\apache-tomcat-7.0.100\conf\.keystore"
Enter keystore password:

Keystore password is too short - must be at least 6 characters
Enter keystore password:

Re-enter new password:

What is your first and last name?
[Unknown]: RUBEN LARA
What is the name of your organizational unit?
[Unknown]: TEC
What is the name of your organization?
[Unknown]: TEC
What is the name of your City or Locality?
[Unknown]: MORELIA
What is the name of your State or Province?
[Unknown]: MICH
What is the two-letter country code for this unit?
[Unknown]: MX
Is CN=RUBEN LARA, OU=TEC, O=TEC, L=MORELIA, ST=MICH, C=MX correct?
[no]: yes

Generating 2,048 bit RSA key pair and self-signed certificate (SHA256withRSA) with a validity of 90 days
for: CN=RUBEN LARA, OU=TEC, O=TEC, L=MORELIA, ST=MICH, C=MX
```

```
<Connector port="8443" protocol="org.apache.coyote.http11.Http11Protocol"
maxThreads="150" SSLEnabled="true" scheme="https" secure="true"
clientAuth="false" sslProtocol="TLS"
keystoreFile="conf/.keystore"
keystorePass="adminadmin" />
```

```
<Connector port="8443" protocol="org.apache.coyote.http11.Http11NioProtocol"
maxThreads="150" SSLEnabled="true">
  <UpgradeProtocol className="org.apache.coyote.http2.Http2Protocol" />
  <SSLHostConfig>
    <Certificate certificateKeystoreFile="conf/.keystore"
certificateKeystorePassword="adminadmin" type="RSA" />
  </SSLHostConfig>
</Connector>
```

Conexión Segura

- JSP (Tomcat)

localhost:8443/WebApplication1/

JSP...

Certificado para RUBEN LARA

Certificado

RUBEN LARA

Nombre del interesado	
País	MX
Estado/Provincia	MICH
Localidad	MORELIA
Organización	TEC
Unidad organizacional	TEC
Nombre común	RUBEN LARA

Nombre del emisor	
País	MX
Estado/Provincia	MICH
Localidad	MORELIA
Organización	TEC
Unidad organizacional	TEC
Nombre común	RUBEN LARA

localhost:8443

cyes

contacto

Certificado para RUBEN LARA

Certificado

RUBEN LARA

Nombre del interesado	
País	MX
Estado/Provincia	MICH
Localidad	MORELIA
Organización	TEC
Unidad organizacional	TEC
Nombre común	RUBEN LARA

Nombre del emisor	
País	MX
Estado/Provincia	MICH
Localidad	MORELIA
Organización	TEC
Unidad organizacional	TEC
Nombre común	RUBEN LARA